

Weather App Project Documentation

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Project Overview

The Weather App is a web-based application that provides real-time weather information from multiple sources, allowing users to compare weather data across different weather services. The application features a responsive design that adapts to weather conditions and time of day, making it both functional and visually engaging.

Technical Implementation

Tools and Technologies Used

- **Frontend Technologies:**
 - HTML
 - CSS
 - JavaScript
- **APIs Integrated:**
 - OpenWeather API
 - WeatherAPI.com
 - Open-Meteo API
 - Frappe Charts for data visualization

Key Features

1. **Multi-Source Weather Data:**
 - Real-time temperature data from three different weather services
 - Comparative visualization through line charts
 - 24-hour forecast
 - 7-day forecast

2. **Dynamic User Interface:**

- Responsive design
- Theme changes based on:
 - Time of day (day/night)
 - Temperature (hot/cold)
- Interactive charts with tooltips

3. **Location Features:**

- City search functionality
- Current location detection
- Favourites system with local storage

4. **Unit Conversion:**

- Celsius
- Fahrenheit
- Kelvin

Implementation Details

• **Local Storage Management:**

- Saves favourite locations.
- Persists last viewed city.
- Maintains user preferences.

• **API Integration:**

- Parallel API calls for real-time comparisons.
- Error handling for failed API requests.
- Data normalization across different sources.

User Manual

Basic Navigation

1. Search for a Location:

- Click the  icon to show the search bar.
- Enter city name and press Enter.
- Or use "Use my location" button.

2. Managing Favourites:

- Click "Set as a Favourite" to save a location.
-  icon indicates favourite status.
- Access favourites through the star icon in navigation.

3. Temperature Units:

- Toggle between Celsius, Fahrenheit, and Kelvin using the dropdown.
- Charts automatically update with unit changes.

Reading the Display

• Main Display:

- Shows current temperature from three sources.
- Weather description and icons.
- Source attribution for each reading.

• Charts:

- 24-hour forecast: Shows temperature trends.
- 7-day forecast: Displays average daily temperatures.
- Hover over data points for detailed information.

Points Achievement (37)

Base Features (26 points)

1. Location Features (3 points)
 - User can search for locations (1 point)
 - Geolocation API implementation (2 points)
2. Weather Data Providers (5 points)
 - Three providers implemented: OpenWeather, WeatherAPI, Open-Meteo (3 + 2 points)
3. Weather Information (7 points)
 - Current weather display (1 point)
 - 24-hour hourly forecast (3 points)
 - 7-day forecast (3 points)
4. Visual Elements (5 points)
 - Weather icons implementation (3 points)
 - Dynamic theming based on weather/time (2 points)
5. Additional Features (6 points)
 - Comparative forecast graphs (3 points)
 - Favourites system (2 points)
 - Temperature unit conversion (1 point)

Additional Project Features (11 points)

1. Well-written PDF report (3 points)
2. Responsive design (4 points)
3. Clear directory structure (1 point)
4. Application works on Firefox, Safari, Edge and Chrome (3 points)

AI Usage Declaration

The following AI systems were used in the development of this project:

1. **Claude AI**

- Used for: Documentation help.
- Specific areas: English language refinement.

2. **ChatGPT**

- Used for: Debugging assistance.
- Specific areas: API error troubleshooting, chart configuration.

All AI-generated code and suggestions were reviewed, modified, and validated to ensure proper functionality and security. The core application logic, design decisions, and implementation strategy were developed independently.